WELEARN GROWTH HUB | LMS PLATFORM INTEGRATION

Edmentum Courseware LTI 1.3 Integration

Technical Onboarding Guide, LTI 1.3 Advantage Registration & Grade Sync (AGS)

Version: 1.0.0 | Date: Nov 2026 | Author: Mani Golzary (Platform Lead) | Target: Edmentum Integration Team

1. Executive Summary & Curriculum Scope

Phase B LMS Roadmap: Embedding accredited courseware into student journeys

WeLearn is expanding its proprietary learning management system, WeLearn Growth Hub, to integrate accredited digital courseware from Edmentum (including Edmentum Courseware grades 6-12 and Apex Learning). This document serves as the formal technical specification and registration dossier for Edmentum's Integration Support team (https://www.edmentum.com/support/integrations/).

Through 1EdTech LTI 1.3 Advantage standards, WeLearn acts as the LTI Platform (Tool Consumer), allowing students to launch assigned Edmentum curriculum modules directly from their 'My Courses' dashboard with single sign-on (SSO), while automated assignment scores, quiz results, and progress metrics return seamlessly into WeLearn's unified teacher gradebook.

COHORT SCOPE & LICENSING POLICY

Edmentum Courseware licenses are strictly allocated to secondary twice-exceptional (W2E) learners and dual-enrolment students taking the Thai National Diploma alongside US high school coursework. Primary learners (WPR) are quarantined from secondary license pools.

2. LTI 1.3 Advantage Architectural Model

WeLearn as LTI 1.3 Tool Platform (Consumer) and Edmentum as Tool Provider

Unlike standard commercial LMS setups (such as Blackboard or Moodle), WeLearn Growth Hub implements the published 1EdTech LTI 1.3 Advantage standard natively on Google Cloud / Firebase serverless microservices. WeLearn fulfills the LMS side of the specification:

- **LTI 1.3 Core Launch:** OpenID Connect (OIDC) 3-legged launch using asymmetric RS256 token signing and JWKS endpoint key verification.
- **Assignment and Grade Services (AGS v2.0):** WeLearn hosts LineItem endpoints where Edmentum automatically posts student scores, attempt states, and timestamps.
- **Deep Linking (v2.0):** Enables WeLearn course coordinators and teachers to browse Edmentum's digital library within the WeLearn course editor and embed specific units.
- **Names and Role Provisioning (NRPS v2.0):** Provides secure class roster synchronization so students are automatically matched to Edmentum sections.

3. LTI 1.3 Registration & Parameter Exchange

Technical endpoints required to establish the trust relationship between WeLearn and Edmentum

To register WeLearn Growth Hub within Edmentum's administrative system as an authorized LTI 1.3 Platform, the following configuration parameters are exchanged bilaterally:

3.1 Parameters Provided by WeLearn (LMS Platform)

Configuration Parameter	WeLearn Value	Purpose & Standards Definition
`Platform ID / Issuer`	`https://growthhub.welearn.org`	Identifies WeLearn as the issuing authority in all JWT claims.
Client ID	(Provided by Edmentum upon setup)	OAuth2 Client ID assigned to WeLearn by Edmentum.
Deployment ID	`wl_deploy_secondary_bkk`	Distinguishes the Bangkok Secondary/W2E school instance.
Public Keyset URL (JWKS)	`https://api.growthhub.welearn.org/lti/jwks.json`	Exposes WeLearn's public RSA keys for signature validation.
OIDC Login Initiation URL	`https://api.growthhub.welearn.org/lti/login_init`	Endpoint where Edmentum initiates the OIDC authentication flow.
`Tool Redirection / Launch URL`	`https://api.growthhub.welearn.org/lti/launch`	Authorized redirect URI receiving the signed LTI launch JWT.

3.2 Parameters Required from Edmentum (Tool Provider)

- **Edmentum OIDC Auth URL:** Endpoint where WeLearn redirects the browser to authenticate the tool session.
- **Edmentum Public Keyset (JWKS):** URI to fetch Edmentum's public keys for verifying Deep Linking and AGS score tokens.
- **Edmentum Access Token URL:** OAuth2 token endpoint (https://.../oauth/token) used for service-to-service authorizations.
- **Target Link URIs:** Direct launch destinations for Courseware dashboard, unit modules, and assessments.

> **PRE-LAUNCH REFERENCE TOOL VERIFICATION**
>
> Prior to connecting live Edmentum production keys, WeLearn's backend is pre-tested and validated against 1EdTech's official LTI Reference Implementation Tool to confirm 100% compliance with Core 1.3 and AGS v2.0 specifications.

4. LTI Deep Linking 2.0 & Content Embedding

Teacher workflow for selecting and placing Edmentum lessons into WeLearn courses

WeLearn course coordinators assemble multidimensional curricula that combine hands-on STEAM projects, solar car simulations, and accredited Edmentum coursework. LTI Deep Linking 2.0 streamlines this process without requiring manual copy-pasting of external URLs:

- **Step 1 (Launch from Editor):** Teacher clicks 'Add Edmentum Module' in WeLearn Course Editor. WeLearn issues a signed LtiDeepLinkingRequest JWT to Edmentum's content selection URL.
- **Step 2 (Content Selection):** Edmentum opens an in-frame picker displaying available courses, units, and quizzes covered by the school's active license.
- **Step 3 (Return Payload):** Upon selection, Edmentum signs an LtiDeepLinkingResponse containing content_items (titles, launch URLs, lineItem score containers, and icons).
- **Step 4 (Course Ingestion):** WeLearn persists the embedded lesson object into Firestore, binding it to the student's 'My Courses' progression tree.

5. Assignment & Grade Services (AGS v2.0) Architecture

Automated synchronization of scores, attempts, and progress back into WeLearn

When a student completes an Edmentum lesson, quiz, or unit post-test, scores pass automatically back to WeLearn Growth Hub via LTI AGS v2.0, eliminating manual grading entry for educators.

Edmentum AGS Field	WeLearn Gradebook Field	Data Type	Processing & Normalization Rule
scoreGiven	rawScore	Decimal Number	Points achieved on assessment (e.g., 85.0).
scoreMaximum	maxPoints	Decimal Number	Maximum possible points (e.g., 100.0).
activityProgress	completionStatus	Enum String	Maps 'Completed' or 'InProgress' to student course progress.
gradingProgress	evaluationState	Enum String	'FullyGraded' triggers immediate visibility in Parent Portal.
timestamp	gradedAt	ISO 8601 UTC	Recorded timestamp of assessment completion.
userId (SisID)	learnerUid	UUID String	Matches student account in WeLearn Firestore multi-tenant schema.

6. WeLearn AGS LineItem Endpoint Specifications

REST contracts hosted on WeLearn backend for Edmentum automated score passback

POST `/api/lti/ags/lineitems/{lineitem_id}/scores`

Called by Edmentum to publish student assessment grades into WeLearn Gradebook.

- Header: Authorization: Bearer <edmentum_access_token>
- Header: Content-Type: application/vnd.ims.lis.v1.score+json
- Scope: https://purl.imsglobal.org/spec/lti-ags/scope/score

```
{
  "timestamp": "2026-11-05T09:15:30Z",
  "scoreGiven": 92.5,
  "scoreMaximum": 100.0,
  "comment": "Unit 3 Mastery Exam: Energy Transformations",
  "activityProgress": "Completed",
  "gradingProgress": "FullyGraded",
  "userId": "wl_stu_98b3f2e1"
}
```

7. Student Identity Mapping & License Governance

SisID provisioning, privacy protection, and license seat management

To ensure compliance with student privacy regulations and prevent license exhaustion, account provisioning adheres to strict rules:

- **SisID Mapping:** WeLearn passes an immutable, pseudonymized learnerUid as the LTI sub / SisID claim. No national ID numbers or birth certificates are exposed.
- **Data Minimization:** LTI launch claims include only educational directory fields: given_name, family_name, email, and roles (Learner or Instructor).
- **License Pool Quotas:** WeLearn maintains an active license counter. Only students enrolled in authorized W2E secondary courses receive launch credentials, preventing primary learners from accidentally consuming secondary seats.

8. Technical Questionnaire for Edmentum Support

Specific items to confirm with https://www.edmentum.com/support/integrations/

Our engineering team requires formal written confirmation from Edmentum technical support on the following 5 integration questions:

- **1. Custom LMS Platform Support:** Does Edmentum support direct 1EdTech LTI 1.3 registration for custom platforms, and what is the approval procedure?
- **2. AGS LineItem Management:** Does Edmentum automatically initiate LineItem creation via AGS v2.0 during launch, or must WeLearn pre-create LineItems via API before student launch?
- **3. Deep Linking Granularity:** Does Edmentum's Deep Linking picker support unit- and lesson-level selections, or only full course-level enrollment?
- **4. Primary Age Products:** Which Edmentum products suitable for primary-age learners (WPR cohort, grades 1-5, e.g., Exact Path / Calvert) are available over LTI 1.3?
- **5. Staging Sandbox Provisioning:** Can Edmentum provision a developer staging account with sample courseware licenses to validate launch and grade returns before pilot deployment?

9. Implementation Milestones & Timeline

Four-week Phase B deployment plan leading to pilot class rollout

Milestone	Duration	Core Deliverable	Success Criteria & Exit Gate
`Phase B.1`	Week 1	LTI Reference Tool Test	100% pass rate on 1EdTech LTI 1.3 Core and AGS automated test suites.
`Phase B.2`	Week 2	Edmentum Registration	Exchange client credentials, JWKS URLs, and deploy staging keys.
`Phase B.3`	Week 3	Deep Linking & Grade Sync	Teacher embeds 3 Courseware units; automated test student scores return to Gradebook.
`Phase B.4`	Week 4	One-Class Pilot (W2E)	20 secondary students launch lessons; verify zero login failures and accurate grade return.

10. Project Leadership & Escalation Contacts

WeLearn technical points of contact for Edmentum integration

- **Platform Lead:** Mani Golzary (AI Learning Architect & IT Platform Lead, mani@welearn.org)
- **Project Sponsor:** David Doran (Founder & CEO, WeLearn Academy)
- **Edmentum Integrations Portal:** https://www.edmentum.com/support/integrations/

ACTION REQUIRED FROM SCHOOL LEADERSHIP

To unblock Phase B execution, school leadership must confirm active Edmentum Courseware license numbers, execute the Edmentum Data Processing Agreement (DPA), and request the LTI 1.3 Platform Registration form from the Edmentum account representative.